Sondow Integrals and Three-Grid Cantor Series

Rigidity, Logarithm Elimination, and Remaining Obstructions

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Abstract

We record an audited continuation of an attempt to prove the irrationality of Euler's constant using Sondow's integrals and a geometric Cantor series with bases $2, 4, 5$. A rational recurrence for the integer coefficient of γ gives a polynomial fractional-part criterion equivalent to irrationality. A family of geometric series consequently has limit 2 in the rational case and lower limit $4/3$ in the irrational case. Further results concern the exact prime-logarithm span, the absence of a fixed-order rational-function recurrence for the integrals, prime-square power structure, and pairwise transcendental differences of normalized integrals. These are auxiliary results: this note does not establish which case holds for γ. Priority for these deductions has not been established by an exhaustive literature review.

1 Definitions and established input

Throughout, log is the real natural logarithm, $n \geq 1$, and

$$d_n = \operatorname{lcm}(1, \ldots, n), \qquad B_n = \binom{2n}{n}, \qquad H_j = \sum_{r=1}^{j} \frac{1}{r}, \quad H_0 = 0.$$

Sondow's integral and its decomposition are [1]

$$I_n = \int_0^1 \int_0^1 \frac{[x(1-x)y(1-y)]^n}{(1-xy)(-\log(xy))}\, dx\, dy, \tag{1}$$

$$I_n = B_n\gamma + L_n - A_n, \qquad A_n = \sum_{i=0}^{n} \binom{n}{i}^2 H_{n+i}, \tag{2}$$

$$L_n = 2\sum_{k=1}^{n} \sum_{i=0}^{\min(k-1,n-k)} \binom{n}{i}^2 (H_{n-i} - H_i) \log(n+k). \tag{3}$$

In particular, $A_n \in \mathbb{Q}$ and L_n is a positive rational combination of $\log(n+1), \ldots, \log(2n)$. Define integers $S_n > 1$, C_n, and M_n by

$$\log S_n = d_{2n} L_n, \qquad C_n = d_{2n} B_n, \qquad M_n = d_{2n} A_n \in \mathbb{Z}.$$

The integer product defining S_n is

$$S_n = \prod_{k=1}^{n} \prod_{i=0}^{\min(k-1,n-k)} \prod_{j=i+1}^{n-i} (n+k)^{(2d_{2n}/j)\binom{n}{i}^2}.$$

With $E_n = d_{2n} I_n$ and $b_n = \{\log S_n\}$, the correct sign convention is

$$E_n = C_n\gamma + \log S_n - M_n. \tag{4}$$

We use Sondow's elementary estimates

$$0 < I_n < 16^{-n}, \qquad d_{2n} < 8^n, \qquad 0 < E_n < 2^{-n}. \tag{5}$$

If $\gamma = u/q \in \mathbb{Q}$, then $q \mid d_{2n}$ for all sufficiently large n. Equation (4) and $0 < E_n < 1$ then give

$$b_n = E_n \quad \text{for all sufficiently large } n. \tag{6}$$

2 A polynomial fractional-part criterion

Lemma 2.1. *Set $R_n = d_{2n+2}/d_{2n}$. Then $R_n \in \mathbb{Z}_{>0}$,*

$$R_n \le 4n + 2, \qquad (n+1)C_{n+1} = (4n+2)R_n C_n.$$

Proof. The least common multiple increases at an integer m only when m is a prime power, in which case its ratio is that prime. The new odd integer $2n+1$ contributes at most $2n+1$; the new even integer $2n+2$ contributes at most 2. This proves the bound. The identity follows from $B_{n+1}/B_n = (4n+2)/(n+1)$. □

Theorem 2.2. *Let $\delta_n = 1/[16(n+1)^2]$. Then*

$$\gamma \notin \mathbb{Q} \quad \Longleftrightarrow \quad b_n \ge \delta_n \text{ for infinitely many } n. \tag{7}$$

Consequently, Sondow's sufficient condition $b_n \ge 2^{-n}$ infinitely often is also necessary for irrationality.

Proof. Put

$$k_n = M_n - \lfloor \log S_n \rfloor \in \mathbb{Z}, \qquad \varepsilon_n = C_n\gamma - k_n = E_n - b_n.$$

Suppose $b_n < \delta_n$ eventually. Since $2^{-n} < \delta_n$ for $n \ge 12$, we have $|\varepsilon_n| < \delta_n$ eventually. Lemma 2.1 gives an integer

$$D_n = (n+1)k_{n+1} - (4n+2)R_n k_n = (4n+2)R_n\varepsilon_n - (n+1)\varepsilon_{n+1}$$

with

$$\begin{aligned} |D_n| &< \frac{(4n+2)^2}{16(n+1)^2} + \frac{n+1}{16(n+2)^2} \\ &< 1 - \frac{15}{16(n+1)} + \frac{1}{4(n+1)^2} < 1. \end{aligned}$$

Thus $D_n = 0$ eventually. Comparing with the recurrence for C_n yields $k_{n+1}/C_{n+1} = k_n/C_n$ eventually. These ratios are one fixed rational number and converge to γ, proving $\gamma \in \mathbb{Q}$.

Conversely, (6) implies $b_n < \delta_n$ eventually if $\gamma \in \mathbb{Q}$. This proves (7). Since $\delta_n > 2^{-n}$ eventually, irrationality implies Sondow's exponential condition infinitely often; the reverse implication follows from (6). □

Corollary 2.3. *If γ is rational then $n^2 b_n \to 0$. If γ is irrational then $\limsup_{n\to\infty} n^2 b_n \ge 1/16$.*

Remark 2.4. This is a rigorously proved dichotomy, not a proof of the infinite-occurrence assertion for the actual sequence S_n. The recurrence argument becomes applicable only after assuming eventual smallness.

3 Consequences for the three-grid series

For a positive direction (a, b, c), follow the ray $t(a, b, c)$ from the origin for $t > 0$. At each positive-integer coordinate plane crossing append 2, 4, or 5 according as the coordinate is x, y, or z. Process simultaneous crossings in the fixed order x, y, z. If the resulting sequence is (q_j), set

$$\Omega(a, b, c) = 1 + \sum_{j \geq 1} \frac{1}{q_1 \cdots q_j}.$$

For any normalized tail of this series, $1/4 \leq T \leq 1$. In particular, after an initial block of m symbols 2, $|\Omega - 2| \leq 2^{-m}$; after an initial block of m symbols 4, $|\Omega - 4/3| \leq (2/3)4^{-m}$.

Theorem 3.1. *Set*

$$a_n = \frac{1}{16(n+1)^3}, \qquad U_n = \Omega(a_n, b_n, E_n).$$

Then

$$\begin{cases} U_n \longrightarrow 2, & \gamma \in \mathbb{Q}, \\ \liminf\limits_{n \to \infty} U_n = 4/3, & \gamma \notin \mathbb{Q}. \end{cases}$$

Proof. By (5), $E_n/a_n < 16(n+1)^3 2^{-n} \to 0$. Before the first z crossing, there are at least $\lfloor a_n/E_n \rfloor$ x crossings. Hence the initial block avoiding the symbol 5 has length tending to infinity. All its symbols are 2 or 4, so comparison with an all-4 block gives $\liminf U_n \geq 4/3$.

If $\gamma \notin \mathbb{Q}$, Theorem 2.2 supplies infinitely many n with $b_n \geq \delta_n$, hence $b_n/a_n \geq n+1$. Along these indices, $E_n < a_n$ eventually and the number of initial y crossings tends to infinity. The initial block consists entirely of 4's, forcing $U_n \to 4/3$ on that subsequence.

If $\gamma \in \mathbb{Q}$, eventually $b_n = E_n$. The number of initial x crossings is then at least $\lfloor a_n/E_n \rfloor \to \infty$, so $U_n \to 2$. The stated tie convention does not affect any of these limits. □

There is also a fixed-gap formulation. For $n \geq 12$ define $V_n = \Omega(\delta_n, b_n, E_n)$. The first symbol is 4 when $b_n > \delta_n$, and 2 when $b_n < \delta_n$, because $E_n < \delta_n$. Equality cannot occur: $\log S_n$, hence b_n, is transcendental by the Hermite–Lindemann theorem. Therefore

$$b_n \geq \delta_n \iff V_n \leq \frac{3}{2}, \qquad b_n < \delta_n \Longrightarrow V_n \geq \frac{13}{8}.$$

Theorem 2.2 makes $V_n \leq 3/2$ infinitely often an equivalent criterion for irrationality. Neither the separation of these ranges nor the limit dichotomy determines the actual branch.

4 What the prime factors prove, and what they do not

Proposition 4.1. *For every odd prime p, there is an integer $T_p > 0$ such that*

$$S_{p-1} = T_p^{p^2}, \qquad v_p(C_{p-1}) = 2, \qquad M_{p-1} \equiv -\frac{d_{2p-2}}{p} \not\equiv 0 \pmod{p}.$$

Proof. The exponent of the displayed base $n + k$ in the product for S_n is

$$e_{n,k} = 2d_{2n} \sum_{i=0}^{\min(k-1, n-k)} \binom{n}{i}^2 (H_{n-i} - H_i).$$

For $n = p - 1$, the summands in $H_{p-1-i} - H_i = \sum_{j=i+1}^{p-1-i} 1/j$ pair under $j \mapsto p - j$. Each pair is $p/[j(p-j)]$, whose denominator is prime to p. Since $v_p(d_{2p-2}) = 1$, every $e_{p-1,k}$ is divisible by p^2. Also $v_p\binom{2p-2}{p-1} = 1$, proving the assertion about C.

Finally, $\binom{p-1}{i}^2 \equiv 1 \pmod{p}$. In $d_{2p-2}H_{p-1+i}$, only the term of denominator p can contribute modulo p, and it occurs exactly for $i \geq 1$. Summing gives $M_{p-1} \equiv (p-1)d_{2p-2}/p \pmod{p}$. □

These congruences do not contradict a small fractional part. Indeed, for any $r \in \{0, \dots, p^2 - 1\}$ and $0 < \eta < 1$, density of $\{j \log 2\}$ supplies $j > 0$ with

$$\frac{r}{p^2} < \{j \log 2\} < \frac{r+\eta}{p^2}.$$

The integer $N = 2^{p^2 j}$ then has $\lfloor \log N \rfloor \equiv r \pmod{p^2}$ and $0 < \{\log N\} < \eta$. This is a counterexample to an inference from power structure and congruences alone, not a counterexample involving Sondow's specific exponents.

5 The exact logarithm space and the recurrence obstruction

Theorem 5.1. *For every $N \geq 1$,*

$$\operatorname{span}_{\mathbb{Q}}\{L_1, \dots, L_N\} = \operatorname{span}_{\mathbb{Q}}\{\log p : p \leq 2N,\ p \text{ prime}\}.$$

Its dimension is $\pi(2N)$.

Proof. Inclusion from left to right follows by factoring the integers in (3). The prime logarithms are linearly independent over $\mathbb{Q}$ by unique factorization. For the reverse inclusion, $L_1 = 2 \log 2$. For an odd prime $p \leq 2N$, put $m = (p+1)/2 \leq N$. In L_m, the coefficient of $\log p$ is positive. All other primes appearing are less than p: the possible bases range from $m+1$ to $2m = p+1$, and the even integer $p+1$ has no prime factor exceeding p. Induction over the primes, or a triangular change of basis, proves the claim. □

Corollary 5.2. *The sequence (I_n) satisfies no nonzero fixed-order homogeneous linear recurrence with coefficients in $\mathbb{Q}(n)$ valid for all sufficiently large n.*

Proof. By (2) and Theorem 5.1,

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{I_1, \dots, I_N\} \geq \pi(2N) - 2 \longrightarrow \infty.$$

A fixed-order recurrence over $\mathbb{Q}(n)$, with leading coefficient not identically zero, expresses all sufficiently late terms as rational linear combinations of finitely many earlier terms. This would bound that dimension independently of N, a contradiction. □

This obstruction concerns the particular sequence (1) and exact recurrences of fixed order. It does not rule out growing-order elimination, recurrences for different approximants, or nonrational coefficients.

6 An unconditional arithmetic consequence for the integrals

Theorem 6.1. *For distinct positive integers m, n, the number*

$$\frac{I_m}{B_m} - \frac{I_n}{B_n}$$

is transcendental. Consequently, among all I_n there is at most one algebraic value. The same at-most-one assertion holds for $E_n = d_{2n} I_n$.

Proof. Write $\alpha_n = A_n/B_n$, $F_n = L_n/B_n$, and $J_n = I_n/B_n$. Let K_n have probabilities $w_n(i) = \binom{n}{i}^2 / B_n$, for $0 \leq i \leq n$. Vandermonde's identity makes these a probability distribution, and $\alpha_n = \mathbb{E} H_{n+K_n}$. For $0 \leq i \leq n$,

$$\frac{w_{n+1}(i)}{w_n(i)} = \frac{B_n}{B_{n+1}} \frac{(n+1)^2}{(n+1-i)^2}$$

increases with i, and the new distribution also has positive mass at $n+1$. The ratios cross 1 at most once, so the cumulative probabilities for K_{n+1} do not exceed those for K_n. Thus K_{n+1} stochastically dominates K_n, and

$$\alpha_{n+1} - \alpha_n \geq \mathbb{E}\frac{1}{n+1+K_n} \geq \frac{1}{2n+1}.$$

Since $F_n = \alpha_n - \gamma + J_n$, estimate (5) gives

$$F_{n+1} - F_n > \frac{1}{2n+1} - \frac{16^{-n}}{B_n} > 0.$$

For $m > n$, $F_m - F_n$ is therefore a nonzero rational linear combination of logarithms of positive integers. Its exponential is a positive algebraic number different from 1. The Hermite–Lindemann theorem implies that $F_m - F_n$ is transcendental [2]. Subtracting the rational number $\alpha_m - \alpha_n$ proves the assertion about $J_m - J_n$. Two algebraic values of I would give an algebraic difference, a contradiction. Multiplication by the nonzero integer d_{2n} preserves algebraicity. □

Remark 6.2. Theorem 6.1 does not distinguish rational from irrational γ. If γ were algebraic, every I_n would already be transcendental by (2), since L_n is a nonzero logarithm of an algebraic number. The theorem is an auxiliary conclusion about the integrals.

7 Exact elimination experiments and the unresolved step

Exact integer arithmetic gives the concrete cancellation

$$\begin{aligned} &- 3923225775E_1 - 158150125E_2 + 392805E_3 \\ &\qquad - 200868E_4 + 35150E_5 = -16097728500\gamma + 8969752951. \end{aligned}$$

All prime-logarithm coefficients cancel exactly. The displayed expression has magnitude approximately 3.22×10^8; this combination does not produce a small integer linear form. Integer-kernel/LLL probes in the windows $[10, 30]$, $[30, 60]$, and $[50, 100]$ likewise produced large forms. These computations test specific reductions, not all possible combinations and not an asymptotic impossibility theorem.

For a growing block of indices, let $\mathcal{L}$ be the integer matrix of prime exponents in $\log S_n$. A vector u in its integer kernel gives

$$\sum_n u_n E_n = \left(\sum_n u_n C_n\right)\gamma - \sum_n u_n M_n.$$

The required assertions are separate:

1. the coefficients must be small enough that $\sum_n |u_n| E_n \to 0$;
2. the resulting form must not vanish identically (or two coefficient pairs must be demonstrably independent).

A nonzero kernel vector need not satisfy the second condition: it can also annihilate the C and M rows. Siegel's lemma alone does not prevent this. The positivity of (1) proves $\int P^2\, d\mu > 0$ for a nonzero polynomial P, but squaring a log-annihilating polynomial need not preserve logarithm cancellation. No estimate resolving these two issues simultaneously was obtained in this attempt.

Current status. The polynomial criterion, geometric dichotomy, prime-power statements, logarithm-span theorem, recurrence obstruction, and normalized-integral transcendence theorem above have proofs. The unconditional assertion $b_n \geq 1/[16(n+1)^2]$ infinitely often remains unproved here. Accordingly, neither the irrationality nor the transcendence of γ has been established by this work.

A The exact tail gcd of the integer coefficients

Proposition A.1. *For $N \geq 1$,*

$$\gcd_{n \geq N} C_n = \begin{cases} 2d_{2N}, & N \text{ is a power of } 2, \\ 4d_{2N}, & \text{otherwise.} \end{cases}$$

The same gcd is obtained from the finite block $N \leq n \leq 2N^2$.

Proof. Fix an odd prime $p \leq 2N$ and $k = \lfloor \log_p(2N) \rfloor$. The lower bound $v_p(C_n) \geq k$ holds for $n \geq N$. Take $m = (p^{k+1} - 1)/2 \geq N$. Its base-$p$ digits are all $(p-1)/2$, so doubling creates no carries and $v_p(B_m) = 0$ by the elementary factorial valuation formula. Also $v_p(d_{2m}) = k$. Thus the gcd has exactly the odd-prime valuations of d_{2N}. Primes greater than $2N$ cannot divide the gcd because they do not divide C_N.

For $p = 2$, the same factorial valuation formula gives

$$v_2(C_m) = \lfloor \log_2(2m) \rfloor + s_2(m),$$

where $s_2(m)$ is the number of binary ones. Set $k = \lfloor \log_2(2N) \rfloor$. If $N = 2^{k-1}$ the minimum is $k+1$, attained at $m = N$. Otherwise every $N \leq m < 2^k$ has at least two binary ones; for $m \geq 2^k$ the minimum is also at least $k+2$, attained at $m = 2^k$. This proves the formula. The odd-prime witnesses satisfy $m < pN \leq 2N^2$, and the binary witness is at most $2N$. □

This narrows the possible denominator if an eventual small-tail assumption has a specified starting index. It does not exclude all denominators when that starting index is unknown.

References

[1] J. Sondow, *Criteria for irrationality of Euler's constant*, Proc. Amer. Math. Soc. **131** (2003), 3335–3344. doi:10.1090/S0002-9939-03-07081-3. Preprint: https://arxiv.org/abs/math/0209070.

[2] A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975. Classical background on exponential and logarithmic transcendence.